REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

# AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

**Course Material: Deep Learning Optimization: SGD, RMSprop & Adam**

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| **Repository:** Fluxbase PostgreSQL • learning_materials | **Vector Store:** Indexed in ChromaDB (notes_collection) |

## 1. Primary Teaching Points & Concept Excerpt

*“Training highly non-convex deep loss landscapes requires adaptive optimization algorithms. While standard SGD with momentum handles ravines, RMSprop and Adam maintain per-parameter adaptive learning rates by tracking running exponential averages of past first and second raw gradients.”*

## 2. Mathematical Formulation & Core Equations

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1. First Moment Vector (Exponential Moving Average):
    m_t = β_1 • m_{t-1} + (1 - β_1) • g_t

2. Second Raw Moment Vector (Variance Proxy):
    v_t = β_2 • v_{t-1} + (1 - β_2) • g_t^2

3. Bias-Corrected Moment Estimators:
    m~_t = m_t / (1 - β_1^t)      v~_t = v_t / (1 - β_2^t)

4. Adam Parameter Update Rule:
    θ_{t+1} = θ_t - ( η / (sqrt(v~_t) + ε) ) • m~_t
```

## 3. 4-Stage Algorithmic Architecture Pipeline

| Stage | Pipeline Component | Mathematical / Architectural Operation | Key Objective |
|---|---|---|---|
| 1 | Gradient Evaluation | Compute mini-batch backpropagation gradient g_t = ∇_θ L | Instantaneous slope |
| 2 | Momentum Accumulation | Update 1st order momentum m_t (β_1 = 0.9) | Dampen oscillations |
| 3 | Curvature Estimation | Update 2nd order uncentered variance v_t (β_2 = 0.999) | Adaptive scaling |
| 4 | Bias Correction & Step | Divide by (1 - β^t) and update parameter weights θ | Normalized parameter update |

## 4. High-Yield Question & Marking Rubric for Midterm IA-2 & SEE Exams

**Expected University Exam Question (8 Marks):** *'Explain why bias correction is strictly required in the Adam optimizer during initial training timesteps. Derive the Adam parameter update rule and compare its behavior against RMSprop on sparse gradients.'*

**Official Faculty Marking Rubric:**

• 2 Marks: Mathematical derivation proving E[m_t] = E[g_t](1 - beta_1^t).

• 3 Marks: Complete step-by-step mathematical update formulation for Adam.

• 3 Marks: Comprehensive analytical comparison table between SGD, RMSprop, and Adam.